

**AIM:**

To analyze telecom customer churn data using Power BI and Tableau by creating interactive visualizations to identify trends in telecom partners, churn, customer age, state, data usage, and call usage.

**Software and Tools Required**

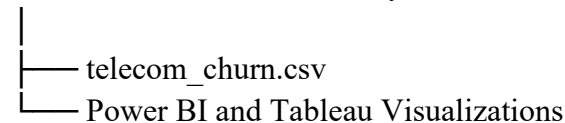
- Power BI Desktop
- Tableau Desktop

**Procedure****Step 1 — Project Setup**

The telecom customer churn dataset was obtained in CSV format and prepared for analysis.

The project uses the following structure:

TelecomCustomerChurnAnalytics

**Step 2 — Import Dataset**

The telecom customer churn dataset was imported into Power BI Desktop and Tableau Desktop for visualization.

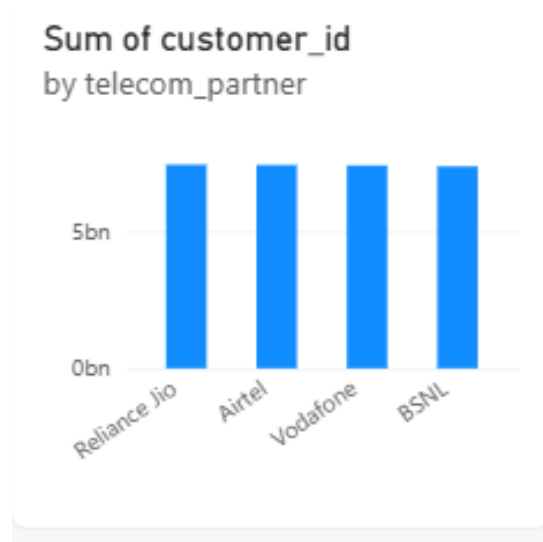
**Data Visualization Using Power BI**

Different visualizations were created to represent telecom customer trends and churn patterns.

**Visualization 1 — Customers by Telecom Partner**

Chart: Column Chart

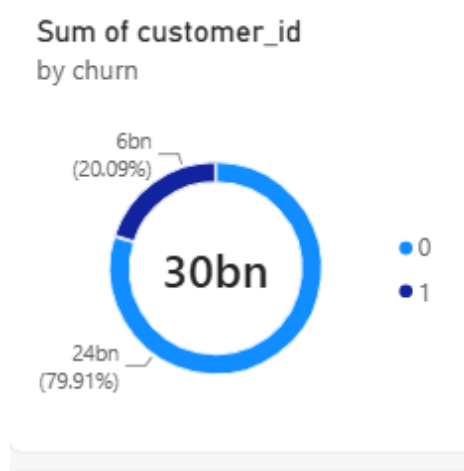
Shows the number of customers for each telecom partner.



## Visualization 2 — Churn Distribution

Chart: Donut Chart

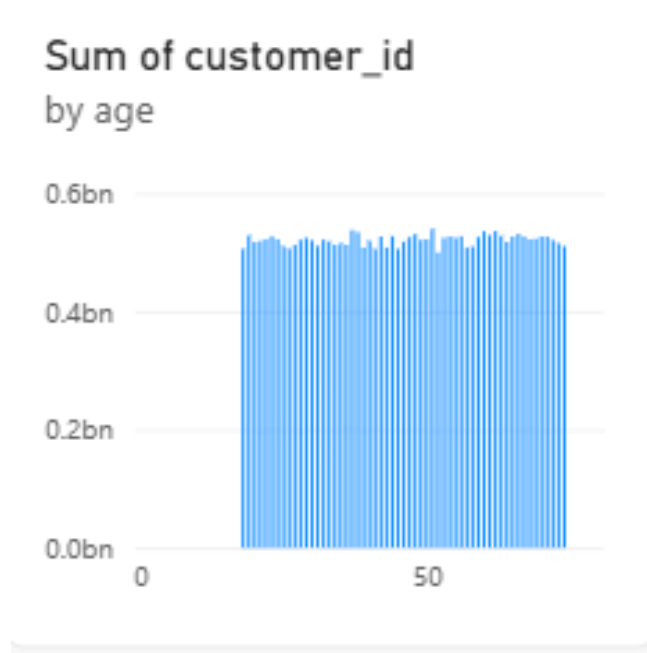
Shows the distribution of customers based on churn.



## Visualization 3 — Customer Age Distribution

Chart: Column Chart

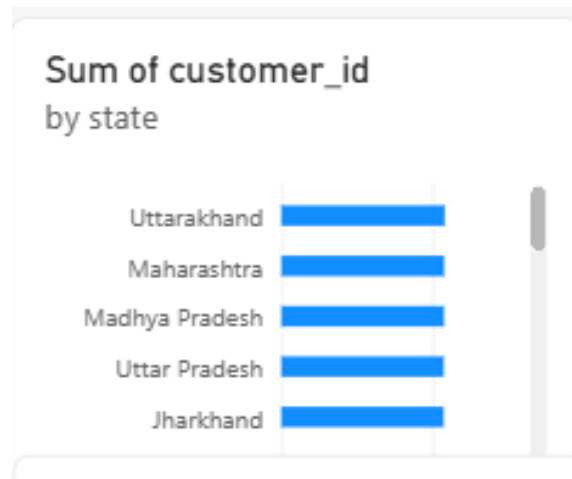
Shows the distribution of customers across different ages.



## Visualization 4 — Customers by State

Chart: Bar Chart

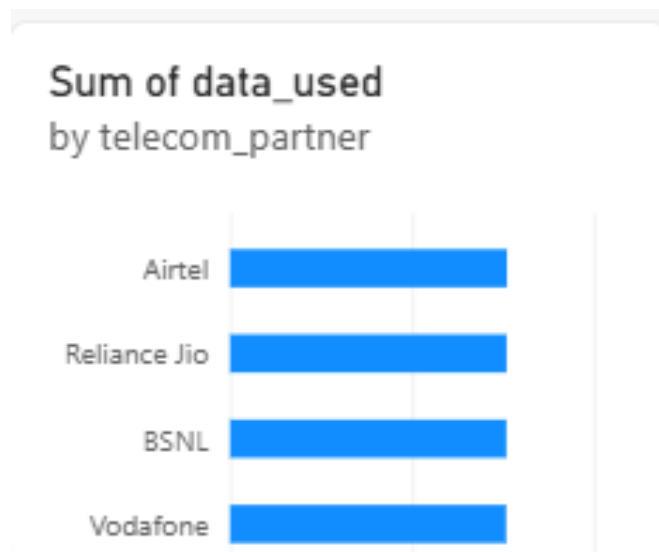
Shows the number of customers across different states.



### Visualization 5 — Average Data Usage by Telecom Partner

Chart: Bar Chart

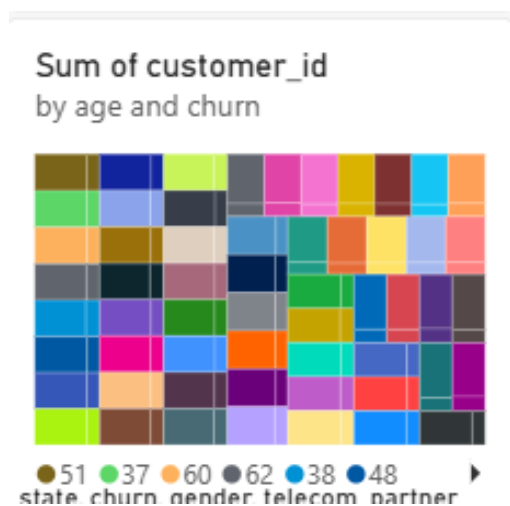
Shows the average data usage for each telecom partner.



### Visualization 6 — Customer Churn by Age

Chart: Treemap

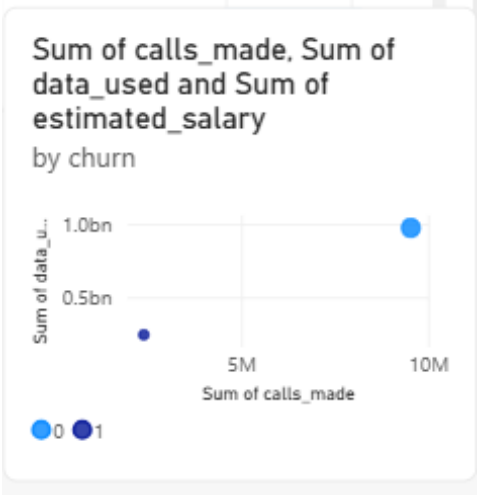
Shows customer distribution by age and churn.



Visualization 7 — Calls vs Data Usage by Churn

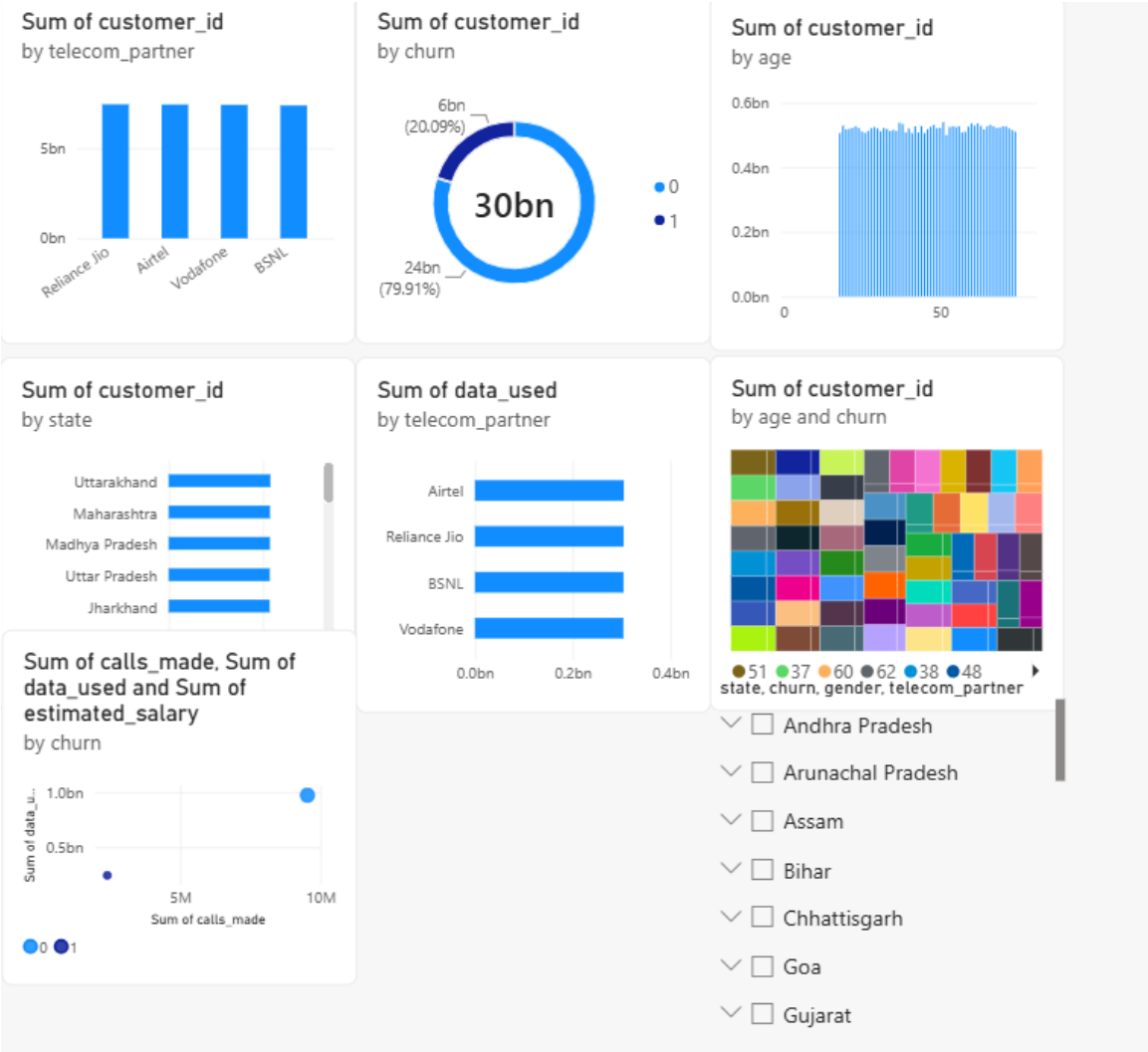
Chart: Scatter Chart

Shows the relationship between calls made and data used based on churn.



OUTPUT:

Power BI Dashboard



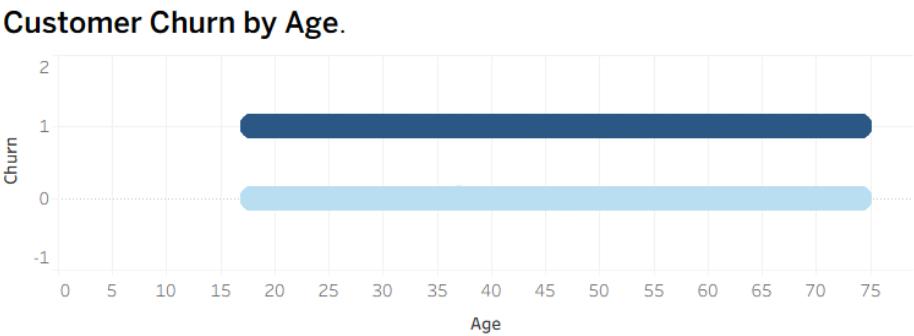
### Data Visualization Using Tableau

Different visualizations were created to represent telecom customer trends and patterns.

#### Visualization 1 — Customer Churn by Age

Chart: Circle Chart

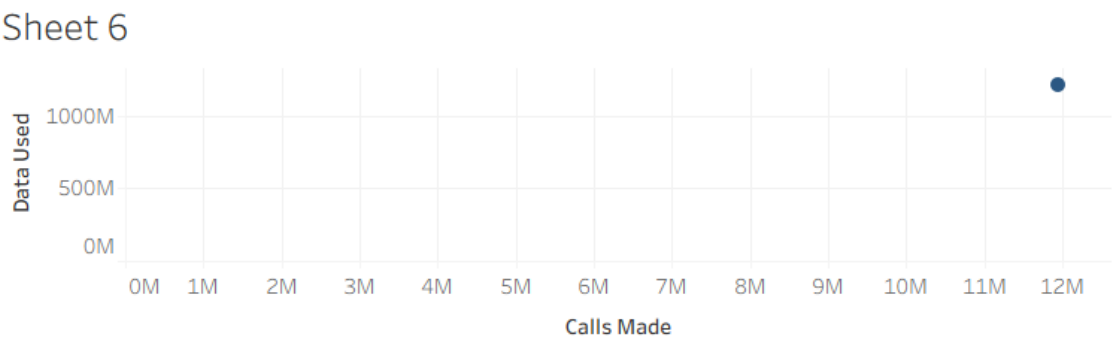
Shows customer churn across different ages.



#### Visualization 2 — Calls vs Data Usage by Churn

Chart: Scatter Chart

Shows the relationship between calls made and data used based on churn.



#### Visualization 3 — Customers by Telecom Partner

Chart: Bar Chart

Shows the number of customers for each telecom partner.

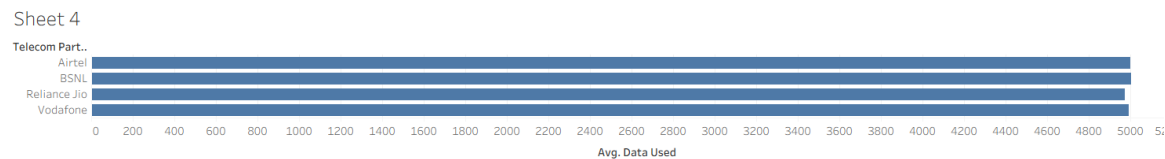
**Customers by Telecom Partner.**

| Customer.. | Telecom Partner |      |          |         |
|------------|-----------------|------|----------|---------|
|            | Airtel          | BSNL | Relian.. | Vodaf.. |
| 1          |                 |      | Abc      |         |
| 2          |                 |      | Abc      |         |
| 3          |                 |      |          | Abc     |
| 4          |                 | Abc  |          |         |
| 5          |                 | Abc  |          |         |
| 6          |                 |      |          | Abc     |

# Visualization 4 — Average Data Usage by Telecom Partner

Chart: Bar Chart

Shows the average data usage for each telecom partner.



## RESULT:

Thus, the telecom customer churn data was successfully analyzed and visualized using Power BI and Tableau. The project helped to identify patterns in telecom partners, customer churn, age, states, data usage, and call usage. The interactive visualizations provide a simple and clear way to understand the telecom customer data.